

SALDEV 1499 Architect Review of Remediation Comment

Flywire Partial sandbox live dry run and implementation quality assessment

Review item	Current evidence
Ticket	SALDEV-1499 Deactivate Quote Line Type background field update flow
Environment	Flywire Partial sandbox Org 00DhG000000jOXUAY
Review date	7 September 2026
Execution result	Focused Apex test run passed 9 of 9
Architect decision	Changes Requested
Boundary	Read-only review. No deployment, activation, metadata edit, Jira comment, or business data change.

Executive decision

The focused Apex class passed all nine tests, and the main structural corrections described in the Jira comment are present. The legacy Flow is inactive, the before-save call order is corrected, and the after-update early return has been replaced with a guarded synchronization block.

Architect approval is still conditional. The current test class does not prove the three exact field outcomes claimed in the comment, and the 200-line test does not provide the stated governor-limit evidence. The correct review disposition is Changes Requested until those tests are strengthened.

Claim outcome

#	Comment claim	Result	Architect finding
1	Before-save order corrected	Pass	Observed in the active trigger source.
2	After-update early return corrected	Pass	Observed as a guarded OLI synchronization block.
3	Three output fields asserted	Fail	The named test asserts only Default_Rate_Details__c is non-null.
4	200-line governor proof	Partial	The test passes, but it lacks the claimed query-limit assertion and mixed-data path.
5	Coverage blocker cleared	Pass	Current focused coverage is 79 percent trigger and 96 percent handler.
6	Flow version 13 deactivated	Pass	The Flow is inactive and version 13 is deactivated.

Evidence reviewed

The Jira screenshots and comment were treated as review evidence, not as instructions to change Salesforce. The live browser session was used only to read the supplied sandbox and execute the focused test class.

Live configuration evidence

Component	Observed state	Review reading
Quote Line Type background field update	Version 13 Deactivated	The legacy Flow is not active.
QuoteLineTrigger	Active API 67.0 modified 7 Sep 2026 01:26	The required trigger structure is present.
QuoteLineTriggerHandler	Active API 67.0 modified 4 Sep 2026 09:10	Bulk Set and Map patterns are used; queries are outside loops.
QuoteLineTriggerHandlerTest	Active API 67.0 modified 7 Sep 2026 05:57	Nine focused tests execute, but assertion quality is incomplete.

Focused dry run

Evidence	Result
Test run ID	707hG00000MwRH0
Completion	Completed 9 total 0 failures
QuoteLineTrigger coverage	79 percent 31 of 39 lines
QuoteLineTriggerHandler coverage	96 percent 280 of 289 lines
Initial attempt	A synchronous attempt met ORG_ADMIN_LOCKED; the same focused class completed asynchronously.

The Developer Console overall coverage display is not used as deployment coverage. The percentages above are the class-level results from the focused run.

Current implementation flow

Order	Trigger path	Observed behavior
1	Before insert and before update	handlePriceChangeOnAmendmentClones
2	Before insert and before update	processBackgroundFields
3	Before insert and before update	copyQuoteLineTypeToBackground
4	After update	OLI synchronization runs only when IDs are available.
5	After update	The transaction can continue to the normal rollup path on the first run.

Validation of the Jira remediation comment

The table below compares each implementation statement with the code and live execution evidence. A passing test run confirms execution, but it does not by itself prove that every asserted business outcome is correct.

Claim	Evidence found	Result	Required follow-up
Before-save sequencing	The trigger calls price-change handling before background processing and the formula-backed copy.	Pass	Keep this order covered by exact outcome tests.
After-update control flow	The early trigger return is replaced by if needsIdSync and quoteIds is not empty. isFirstRun is restored in finally.	Pass	Add one test that proves OLI sync and rollup both occur in the same update.
Output field assertions	testProcessBackgroundFieldsFallbackAndEst oreMismatch queries only Default_Rate_Details__c and checks non-null.	Fail	Assert exact values for all three claimed fields after re-query.
200-line proof	testBulk200QuoteLinesNoSql101Error inserts 200 lines, disables managed triggers for insert, then calls calculateRollupField directly.	Partial	Exercise the complete trigger path and record governor headroom and outcomes.
Coverage	Focused run completed with 79 percent trigger coverage and 96 percent handler coverage.	Pass	Retain the run ID in the review evidence.
Flow deactivation	Flow list shows inactive and Flow Builder identifies version 13 as deactivated.	Pass	Do not reactivate the Flow as part of this fix.

Decision basis

Items 1, 2, 5, and 6 are verified. Item 3 is not substantiated by the current test source. Item 4 is only partially substantiated because the named method completes without an exception but does not demonstrate the explicit query-limit assertion or end-to-end bulk behavior stated in the comment.

Technical findings

Priority	Finding	Impact	Required correction
High	Exact business outcomes are not asserted	A green test can coexist with incorrect Quote Line Type background, rate-detail mismatch, or fallback values.	Re-query and assert the exact expected value of Quote_Line_Type_background__c, Rate_Details_Changed__c, and Default_Rate_Details__c.
High	Formula copy correctness after price-change updates is unproven	Quote_Line_Type__c is a formula read before its value is copied to Quote_Line_Type_background__c; the test does not prove the formula reflects the final in-transaction state.	Add a scenario that changes Price_Change__c and proves the persisted background classification after the transaction.
Medium	Bulk claim is overstated	The 200-line method uses one quote and one product, bypasses managed insert triggers, calls the handler directly, and does not assert query count.	Use mixed source, subscription, amendment, and quote data through the complete supported path; assert final results and defensible limit headroom.
Medium	Combined OLI synchronization and rollup behavior lacks regression proof	The structural early-return fix is visible, but no behavioral assertion proves both downstream actions occur in one update.	Add a focused update test that validates the OLI result and the rollup result together.
Medium	Generic exception handling can hide classification failures	copyQuoteLineTypeToBackground catches a generic exception and only logs a warning.	Narrow the caught condition or surface a failure that tests and operations can detect.
Low	Test method name differs from the Jira comment	The code uses EstoreMismatch rather than RestoreMismatch, which makes evidence mapping harder.	Align the method name and the Jira evidence when the test is revised.

Best practice assessment

Practice	Assessment	Finding
Bulkification	Pass	Set and Map collection is used, with SOQL outside loops.
Trigger orchestration	Pass	Before-save sequencing is explicit and after-update synchronization is guarded.
Recursion safety	Pass	isFirstRun is restored with try and finally before the normal recursion guard.
Assertions	Fail	Current assertions do not prove all business outputs named in the remediation comment.
Governor proof	Partial	The test handles 200 records but does not provide the claimed limit measurement or realistic mixed transaction.
Failure visibility	Partial	A generic catch can allow classification failure to pass with only a warning.

Required actions before approval

The implementation can be re-reviewed after the following evidence is added. These changes should remain limited to SALDEV-1499 and its focused regression coverage.

#	Required action	Acceptance evidence
1	Assert all three output fields	Exact expected values after insert and update for background type, rate-detail changed flag, and default rate details.
2	Strengthen the 200-line transaction	Mixed business scenarios, complete trigger route, no SOQL-101 failure, final field and rollup assertions, and recorded limit headroom.
3	Prove OLI synchronization and rollup together	One update demonstrates that both downstream behaviors complete after the guarded synchronization block.
4	Prove formula-backed copy behavior	A Price_Change__c scenario confirms the persisted background classification matches the final formula result.
5	Improve failure handling	Expected exceptional cases are handled narrowly; unexpected failures are visible to tests and support.

Approval conditions

- The focused test class passes after the assertion changes.
- The exact field values are documented for each covered scenario.
- The bulk evidence uses a representative transaction and records governor headroom.
- The OLI synchronization and existing rollup behavior pass in the same update.
- The Flow remains deactivated and no unrelated automation is changed.

Review conclusion

Execution status is Pass. Architect review status is Changes Requested. The implementation direction is sound, but the current automated evidence does not support two material claims in the Jira comment. Approval should follow the strengthened tests and a new focused dry run.

Ready to paste architect response

Architect review completed for SALDEV-1499 in the Flywire Partial sandbox. The focused Apex run 707hG00000MwRH0 completed with 9 of 9 tests passing, 79 percent coverage for QuoteLineTrigger, and 96 percent coverage for QuoteLineTriggerHandler. The Flow is deactivated, the before-save order is corrected, and the after-update early return is replaced with a guarded OLI synchronization block.

Changes Requested. The Jira comment overstates the automated proof. testProcessBackgroundFieldsFallbackAndEstoreMismatch does not assert Quote_Line_Type_background__c or Rate_Details_Changed__c and only checks that Default_Rate_Details__c is non-null. testBulk200QuoteLinesNoSoql101Error completes for 200 records, but it does not assert Limits.getQueries below 100, uses one quote and one product, disables managed triggers for the insert, and calls calculateRollupField directly. Add exact field assertions, representative end-to-end bulk evidence, a combined OLI-sync and rollup test, and a formula-copy scenario before approval.

This was a read-only review. No deployment, activation, metadata edit, Jira comment, or business data change was performed.

Final recommendation

Area	Status	Recommendation
Structural remediation	Pass	Retain the current direction.
Focused execution	Pass	Keep run 707hG00000MwRH0 as evidence.
Business outcome proof	Fail	Add exact assertions before approval.
Bulk and governor proof	Partial	Replace the narrow direct-handler test with representative end-to-end evidence.
Promotion readiness	Fail	Changes Requested until the evidence gaps are closed.